Simulation of C-D nozzle flow and Transient Radial Heat Conduction

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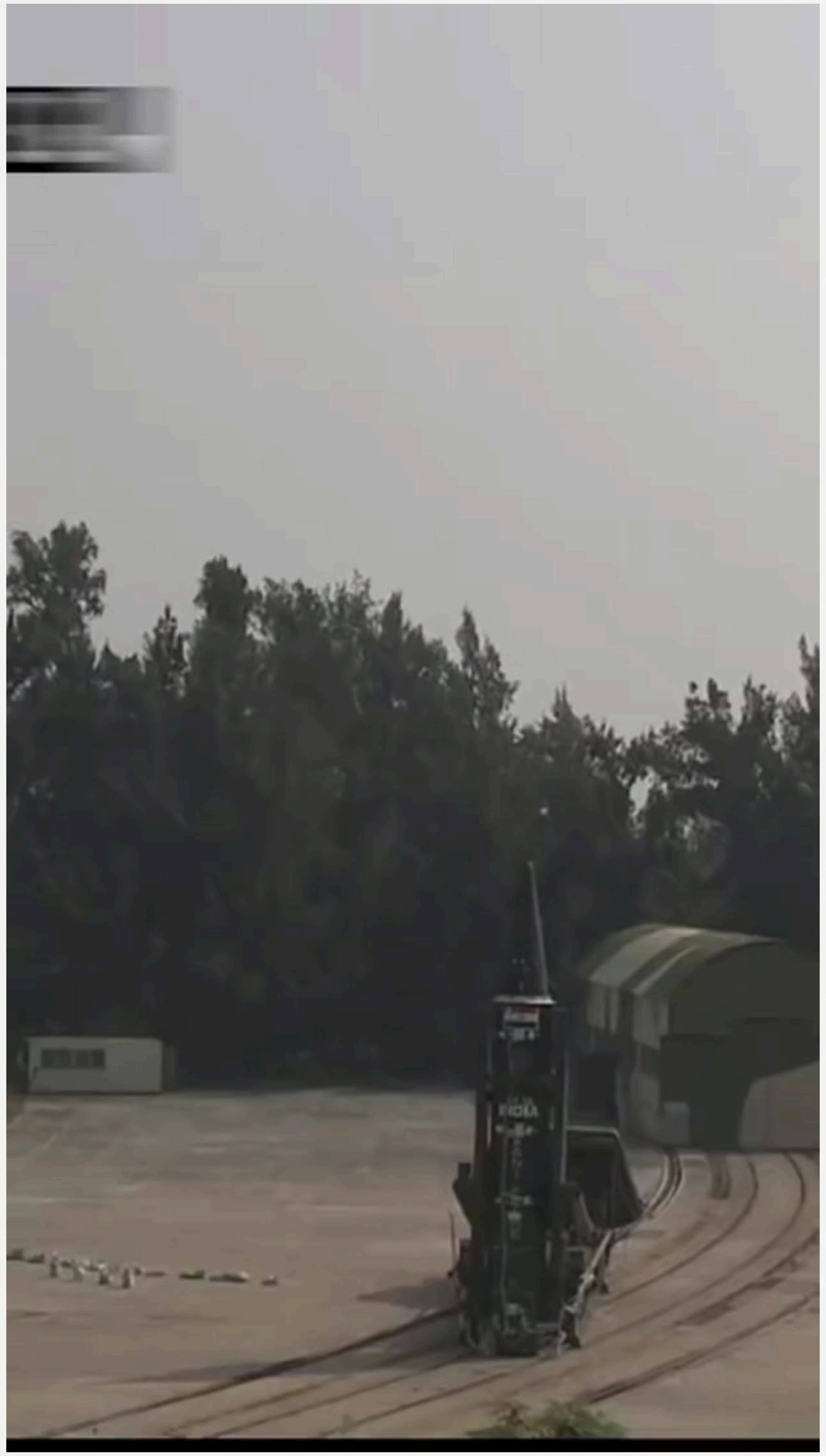
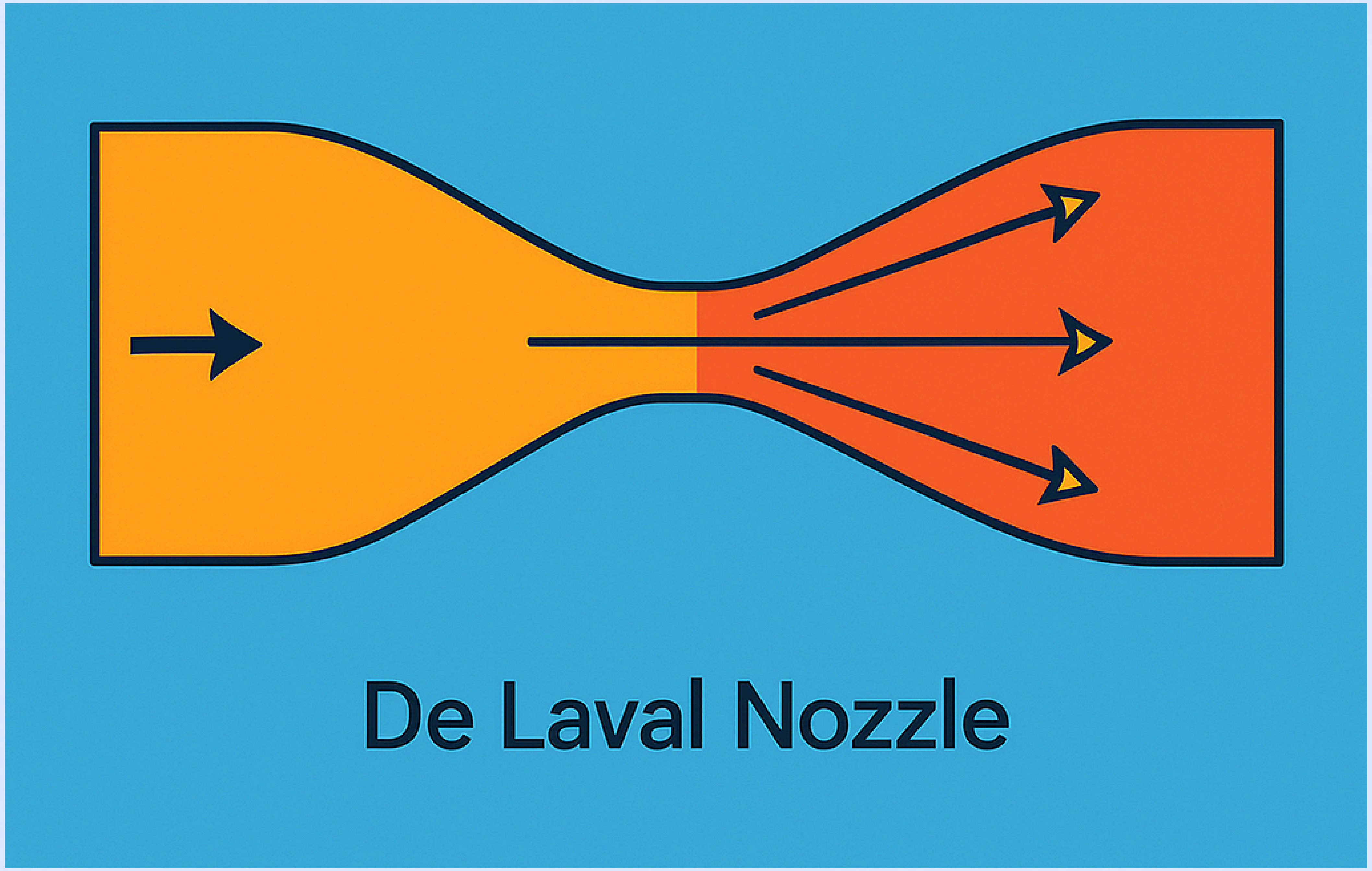


Defence Research and Development Organisation (DRDO)

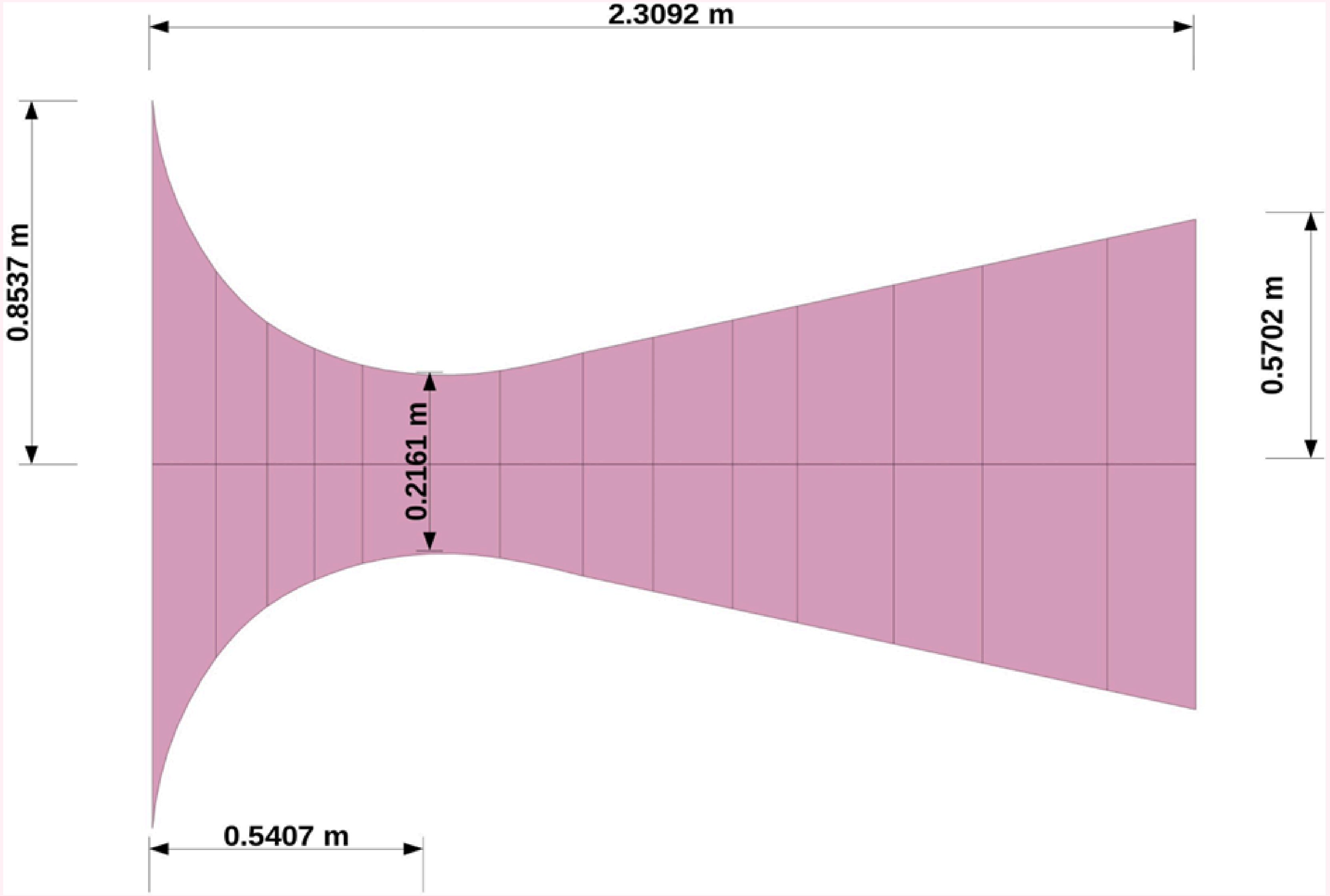
- DRDO strengthens India’s defence through innovation and self-reliance.
- Developed Brahmos, Agni, Prithvi, Tejas, Pinaka, Akash, Abhyas, radars, EW systems, etc.
- Expanded from 10 labs to 41 labs + 5 DYSLs.
- Equips armed forces with advanced weapons and critical defence tech.

Simulation of C-D Nozzle flow



Problem Statement

- To study Mach number variation in a C-D nozzle.
- Python solver (area–Mach relationship).
- Used CFD++ software for simulation.
- Inlet conditions: 40 bar, 2800 K, 10 m/s.
- Wall at 300 K.
- Validate Python results with CFD++ output.



Python Code

Area ratio-Mach number Empirical relationship:

$$\frac{A}{A_{throat}} = \frac{1}{M}\left[\frac{2}{\gamma + 1}\left(1 + \frac{\gamma - 1}{2}\right)M^2\right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

Representing the Empirical Relationship as a function:

$$f(M) = \frac{1}{M}\left[\frac{2}{\gamma+1}\left(1 + \frac{\gamma-1}{2}\right)M^2\right]^{\frac{\gamma+1}{2(\gamma-1)}} - \frac{A}{A_{throat}}$$

Bisection Method:

If solution of f(x) lies in [a,b]

Calculate $\frac{a+b}{2}$

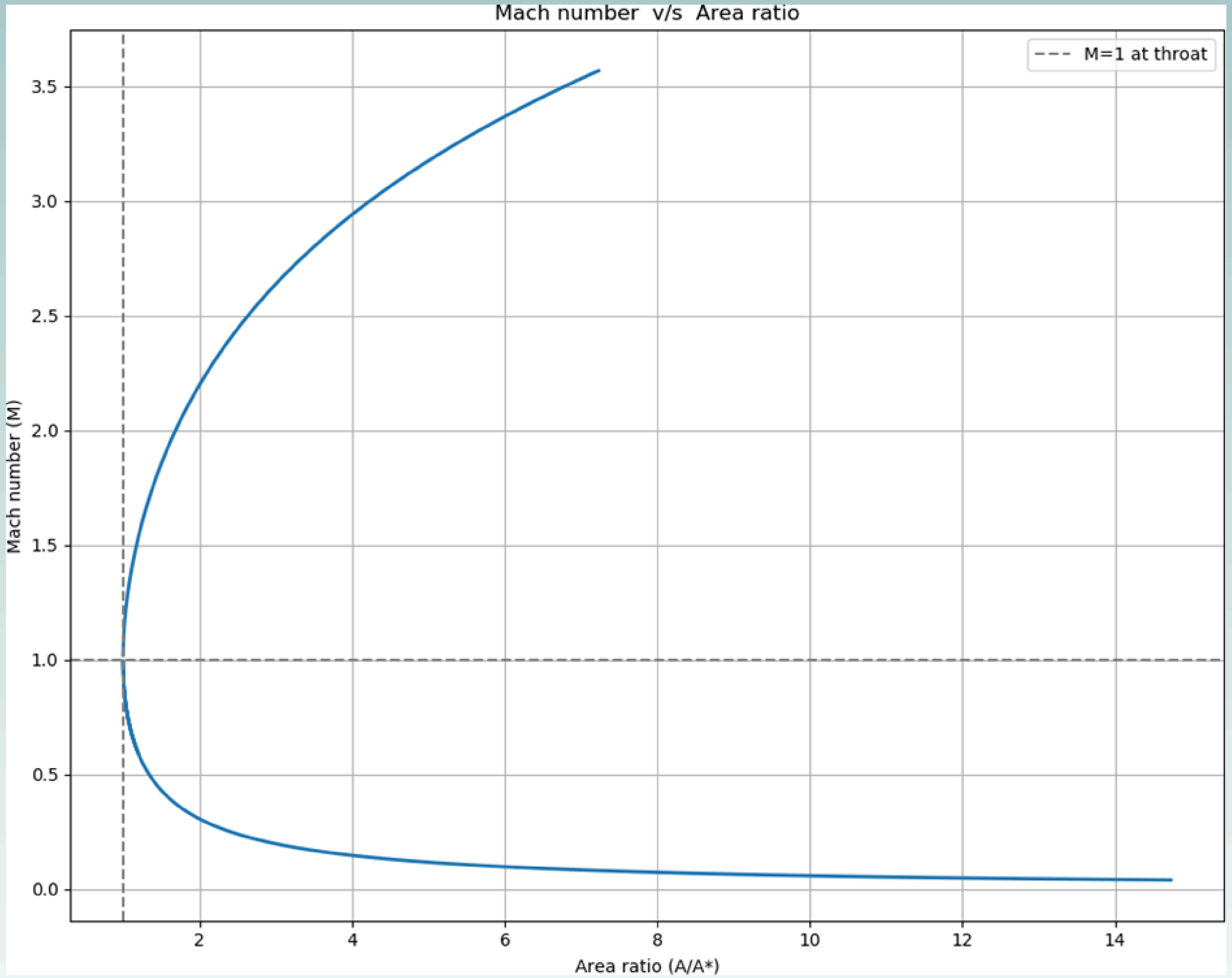
If $f(a) \cdot f\left(\frac{a+b}{2}\right) < 0$

If $f\left(\frac{a+b}{2}\right) \cdot f(b) < 0$

Results : Python Code

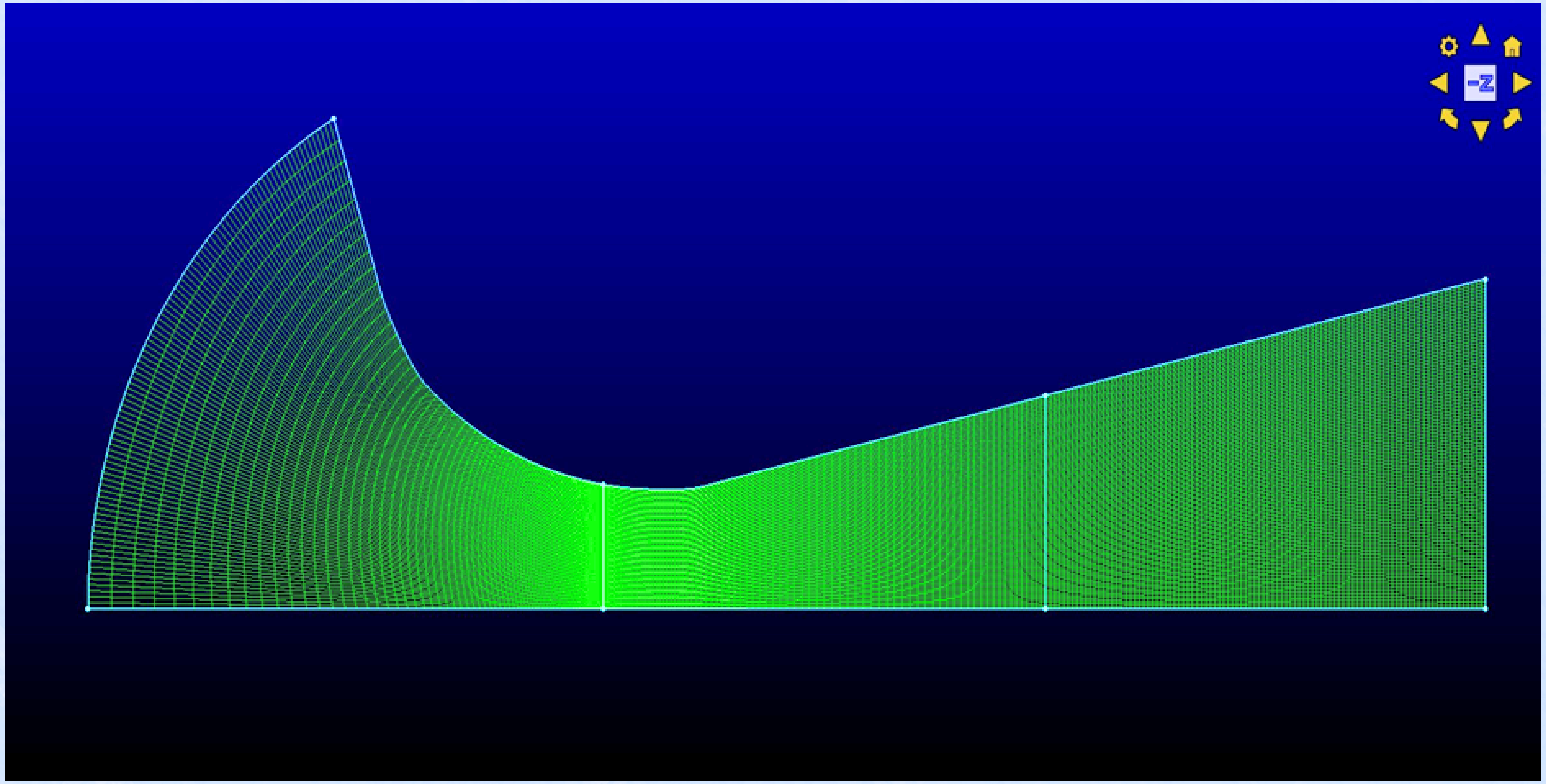
Maximum Mach number : 3.567

Minimum Mach number : 0.0393



CFD++ Simulation

Meshing in Pointwise



CFD++ Simulation

Boundary Conditions names:

- Inlet
- Wall
- Outlet
- Symmetry

Inlet Conditions:

- Pressure, P = 40 bar
- Velocity, v = 10 m/s
- Temperature, T = 2800K

Wall temperature = 300K

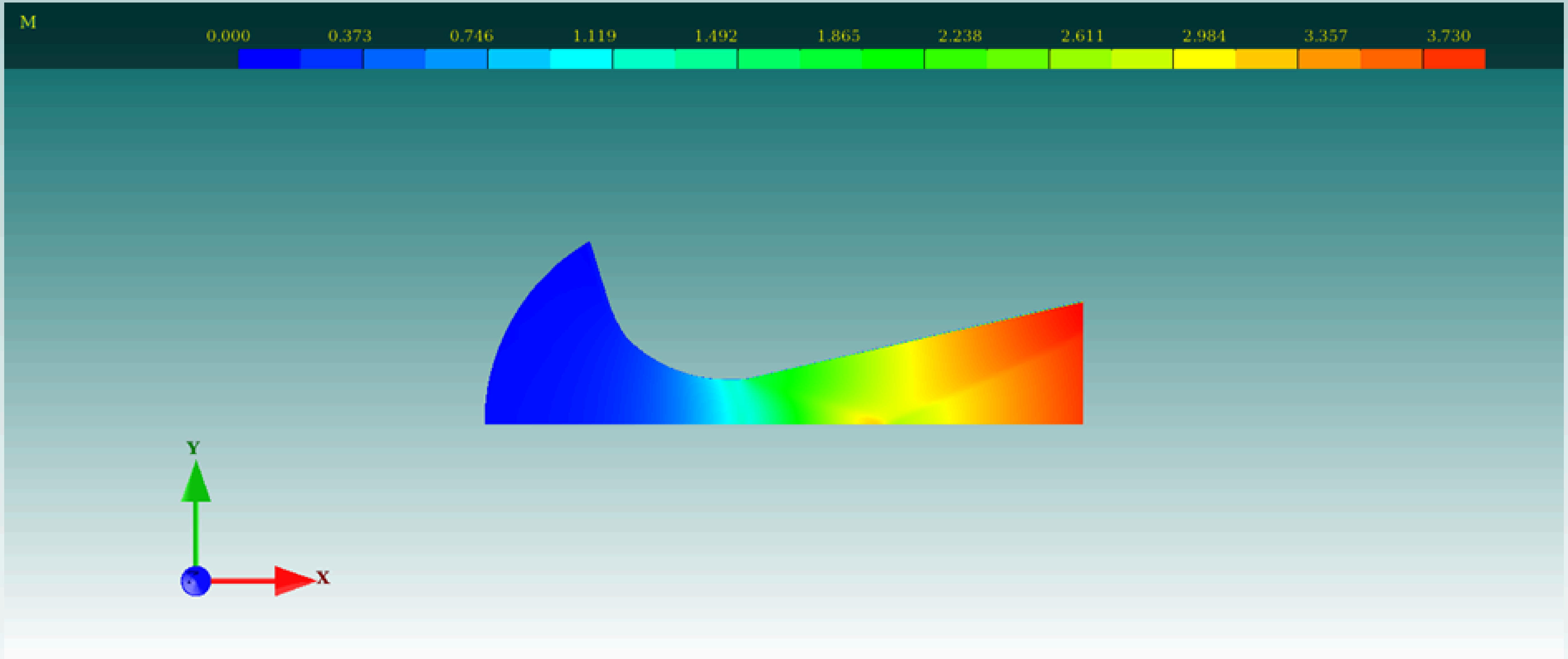
Outflow : Supersonic

Results : CFD++

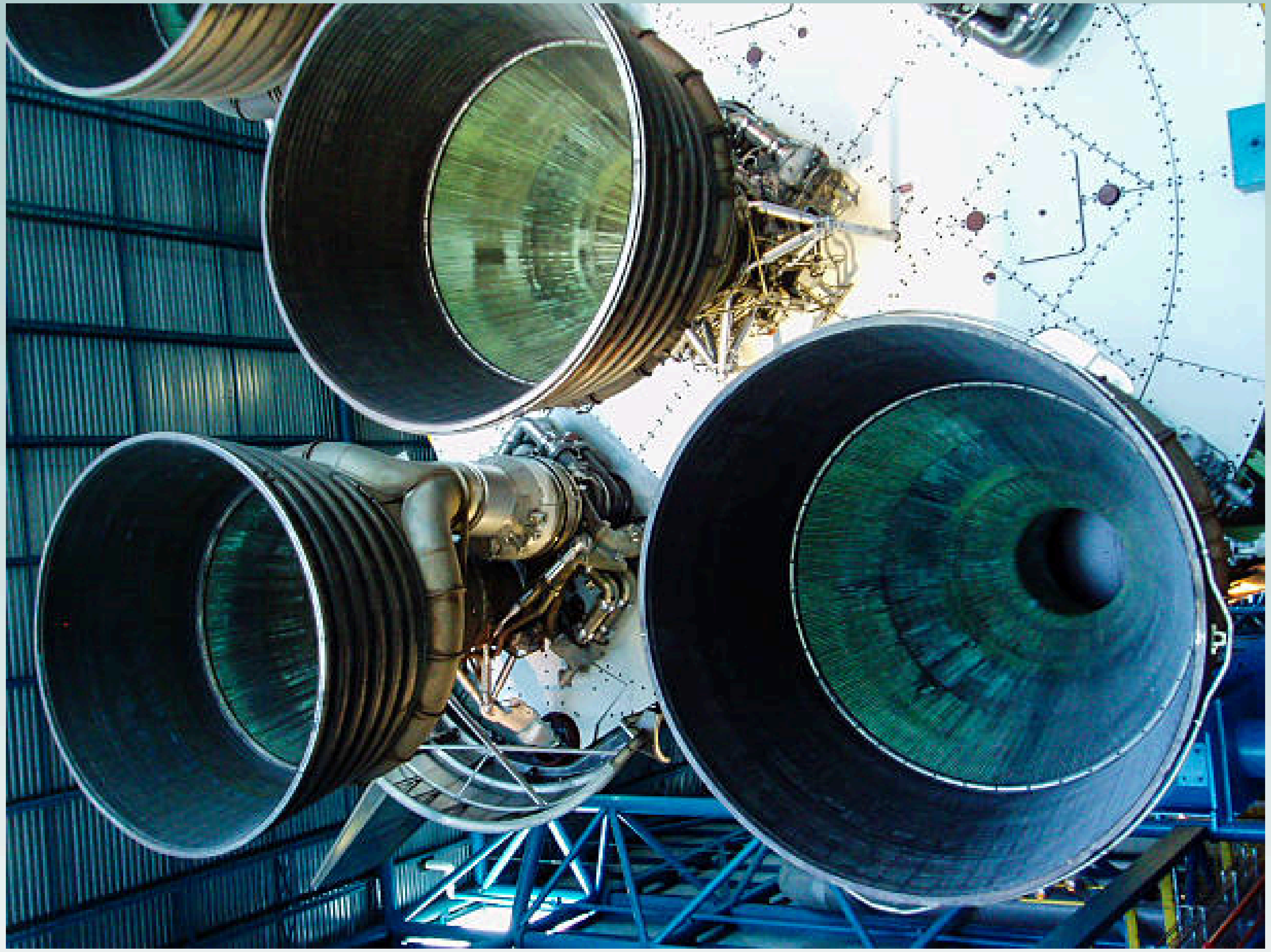
Contour plot of Mach number:

Maximum Mach number : 3.7296

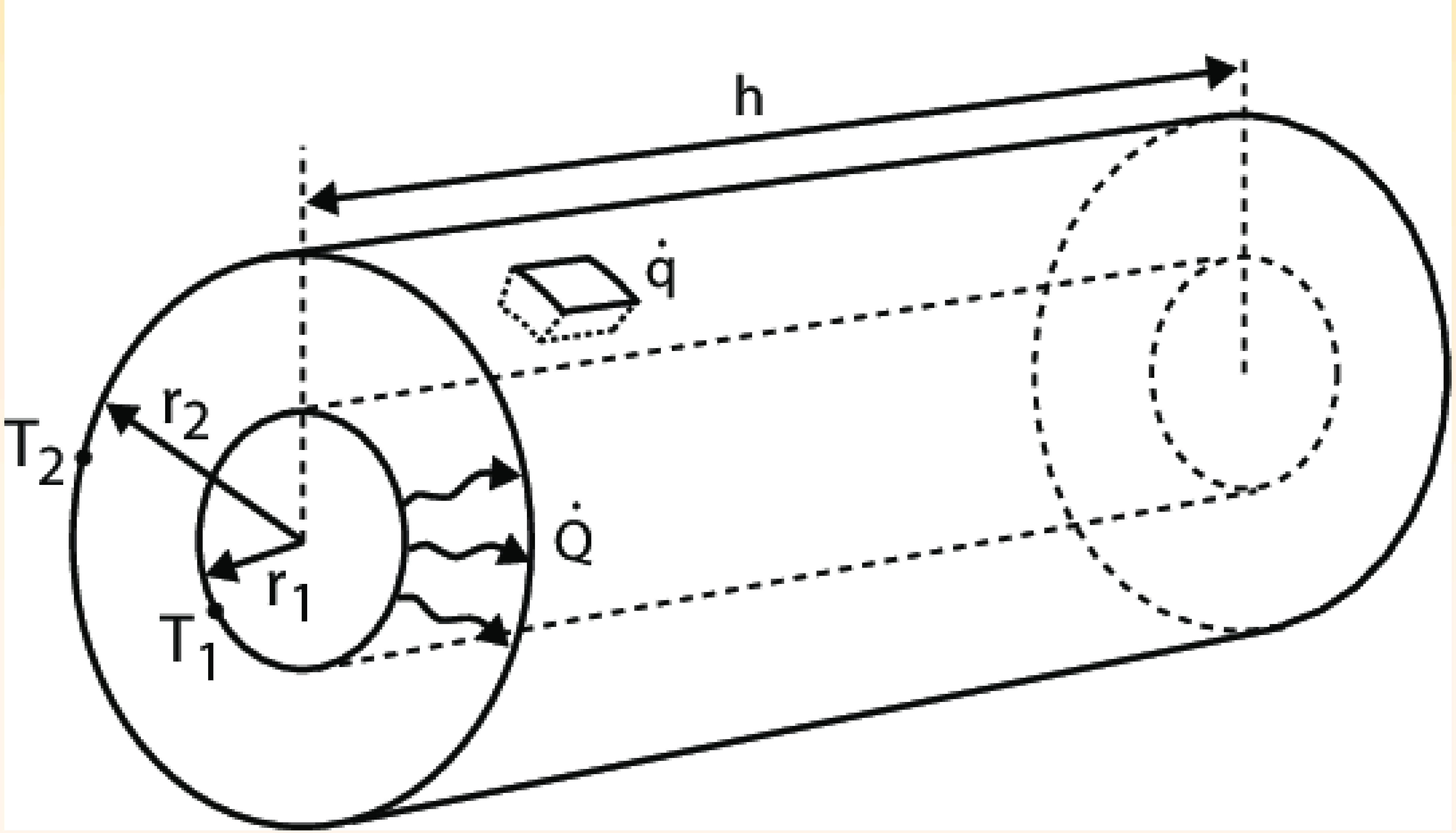
Minimum Mach number : 0.0



C-D Nozzle in Rocket Propulsion



Transient Radial Heat Conduction



Problem Statement

- Annular sector was taken.
- Solved with Python (FVM + implicit scheme, TDMA) for stability and efficiency.
- Geometry: $r_{inner} = 2.005$ m, $r_{outer} = 2.010$ m.
- Properties: $\rho = 1400$ kg/m^3, $C_p = 1000$ J/kgK, $k = 0.1$ W/mK, $T_0 = 300$ K.
- Boundary: $q_{inner} = 20{,}000$ W/m^2, $q_{outer} = 0$ W/m^2.
- Verified with Ansys APDL under identical conditions.

Python Code

Material Properties:

- Specific heat, C = 1000 J/kg·K
- Density, ρ = 1400 kg/m³
- Thermal conductivity, k = 0.1 W/m·K

Transient Radial Heat Conduction Equation:

$$\rho C_p \frac{dT}{dr} = \frac{1}{r}\frac{d}{dr}\left(kr\frac{dT}{dr}\right)$$

TDMA (Tri-Diagonal Matrix Algorithm):

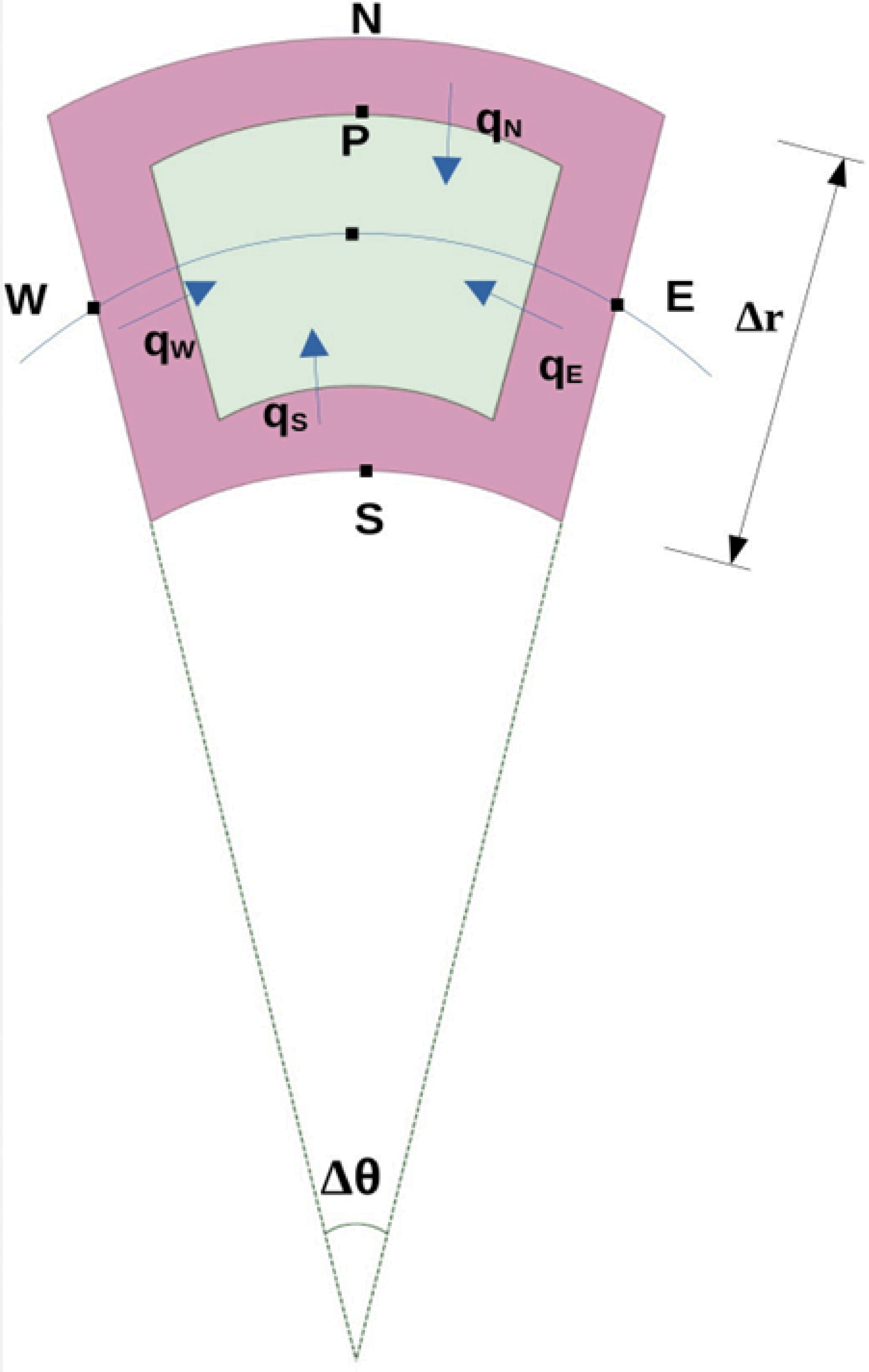
A X = B

A : Tridiagonal Matrix
X : Variable Matrix
B : Constants Matrix

Finite Volume Method

Taking a cell in which the green area is the control volume,

- P : Present cell
- N : North cell
- S : South cell
- W : West cell
- E : East cell



Ansys APDL Simulation

Element type:
Quad 4 Node 55
Included angle:
6 degrees
(0.1 radians)

q_{inner} = 20,000 W/m^2
Initial condition
Temperature = 300K

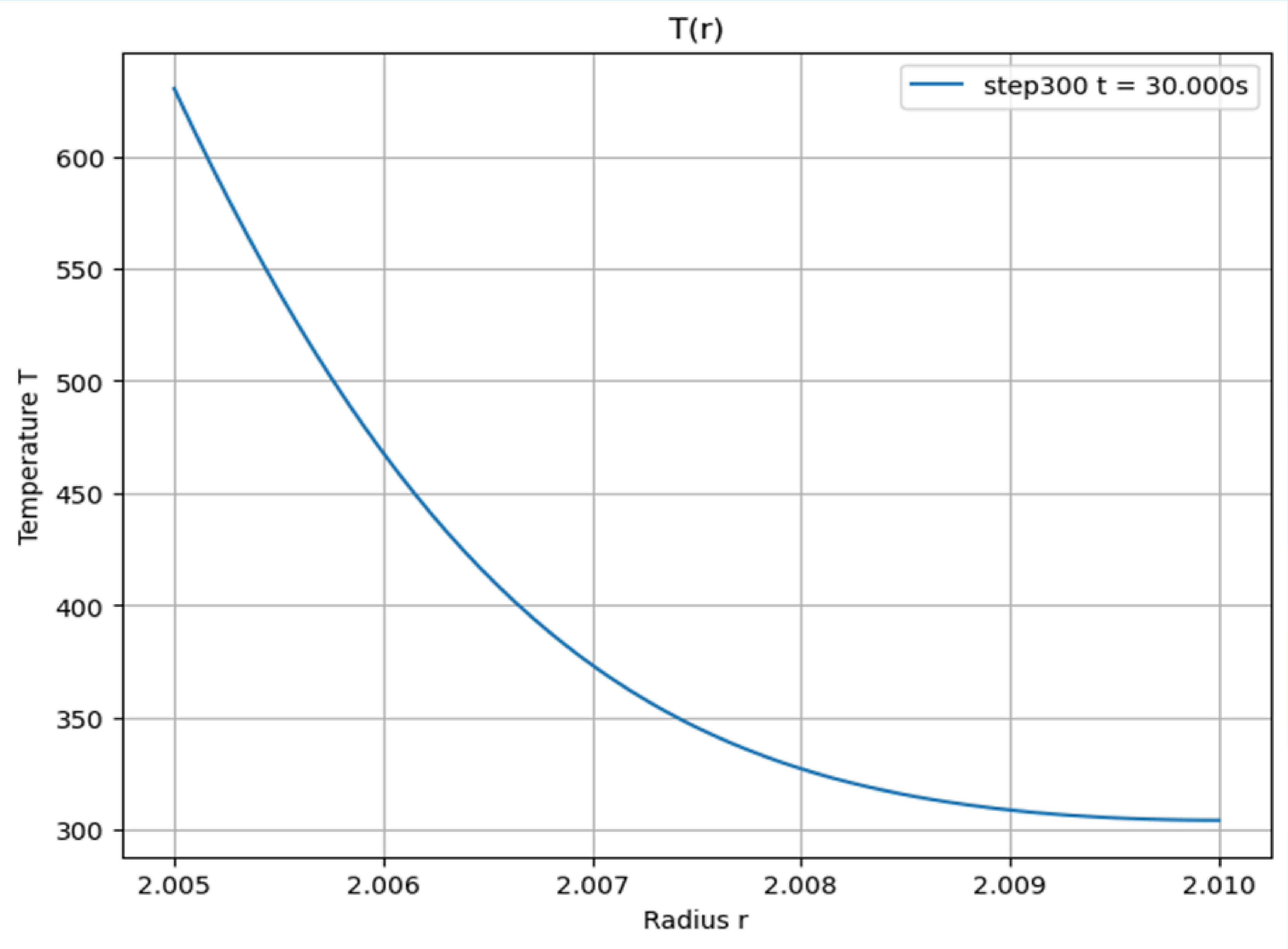
Inner radius:
2.01 m
Outer radius:
2.005

Simulation time:
30 seconds

Results : Python code

Temperature profile of annular sector after 30 seconds

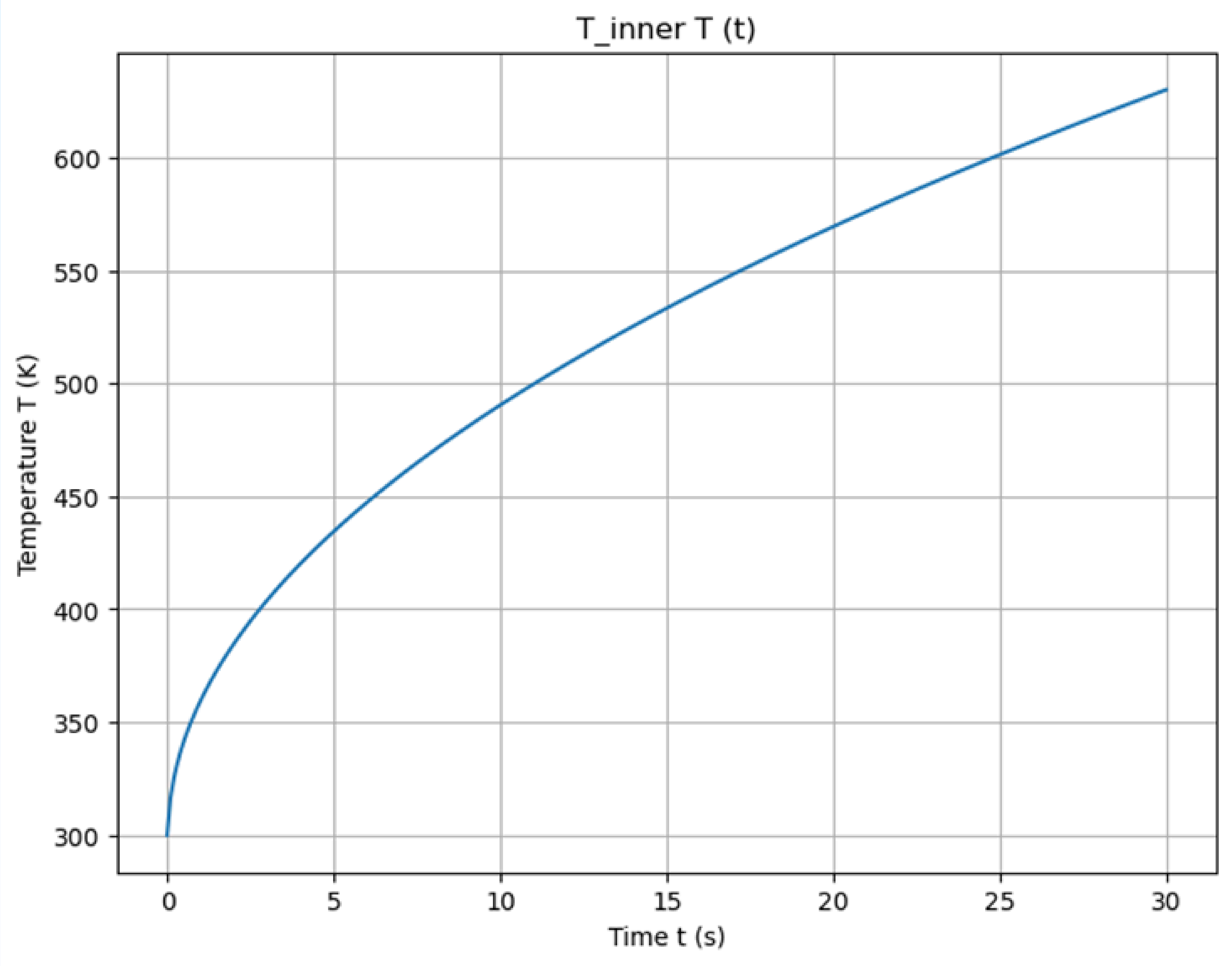
- Inner wall Temperature = 630.0868 K
- Outer wall Temperature = 304.3718 K



Results : Python code

Temperature variation of inner cell:

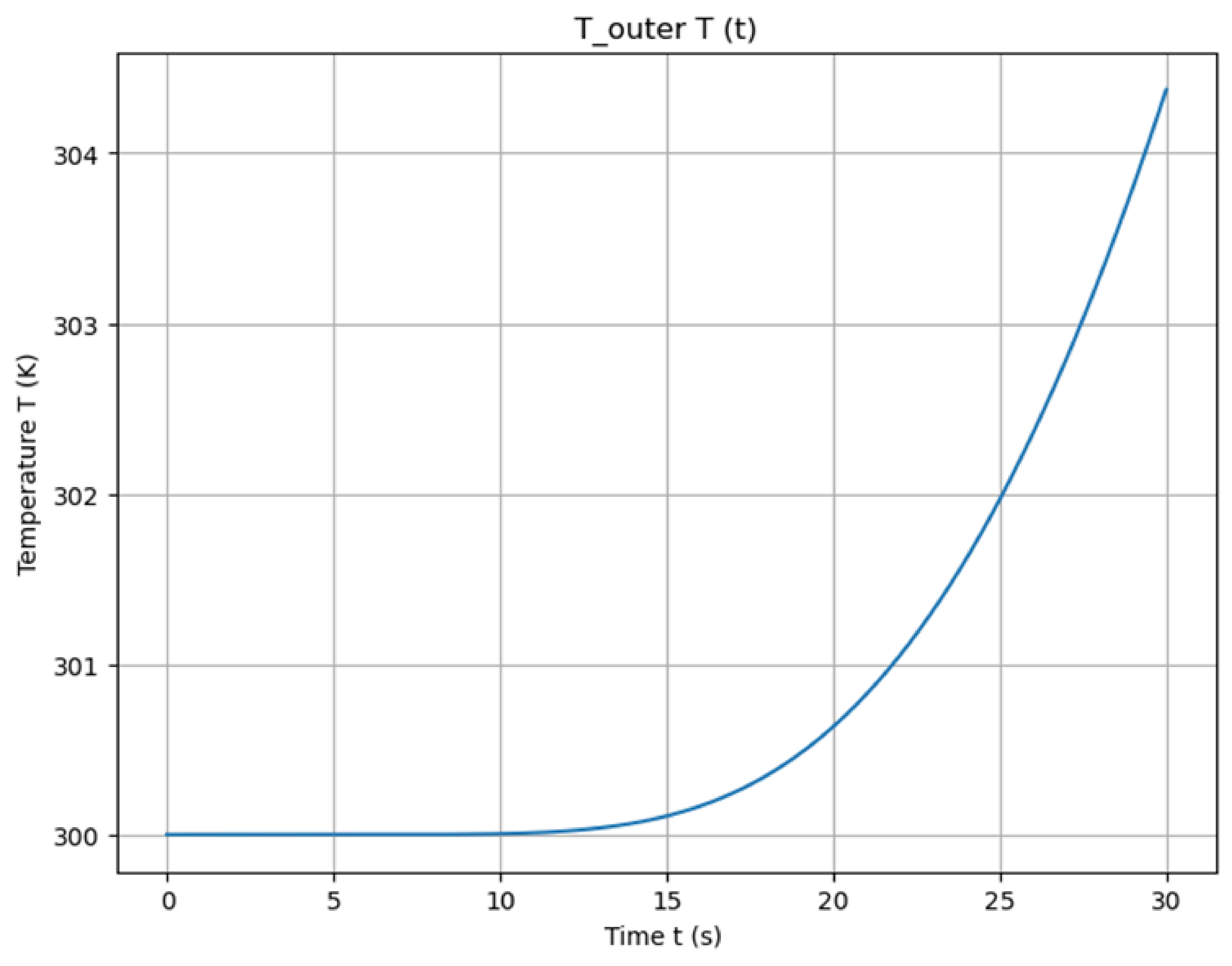
Maximum Temperature: 630.0868 K



Results : Python code

Temperature variation of outer cell:

Maximum Temperature: 304.3718 K.

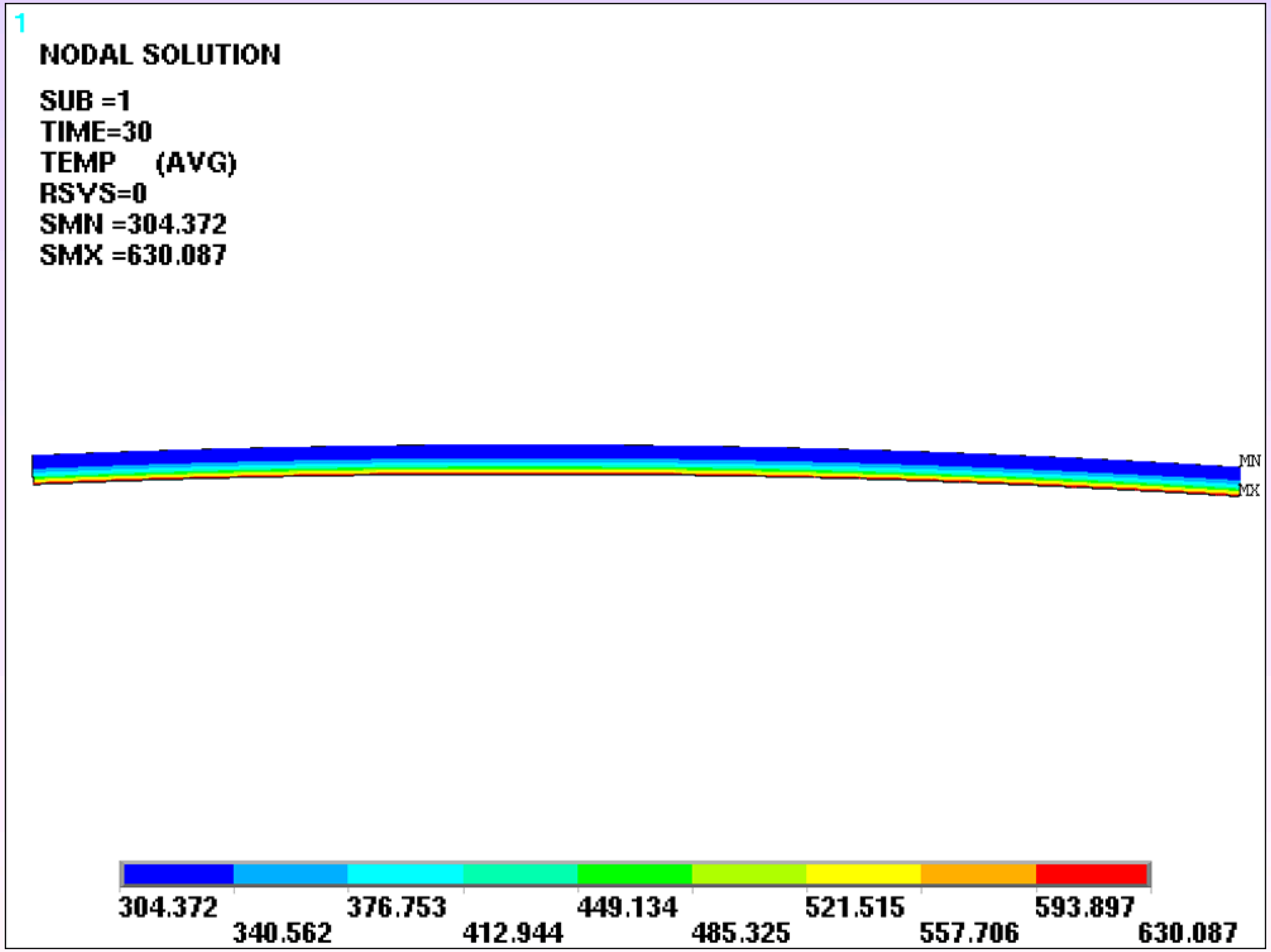


Results : Ansys APDL

Contour plot of annular sector:

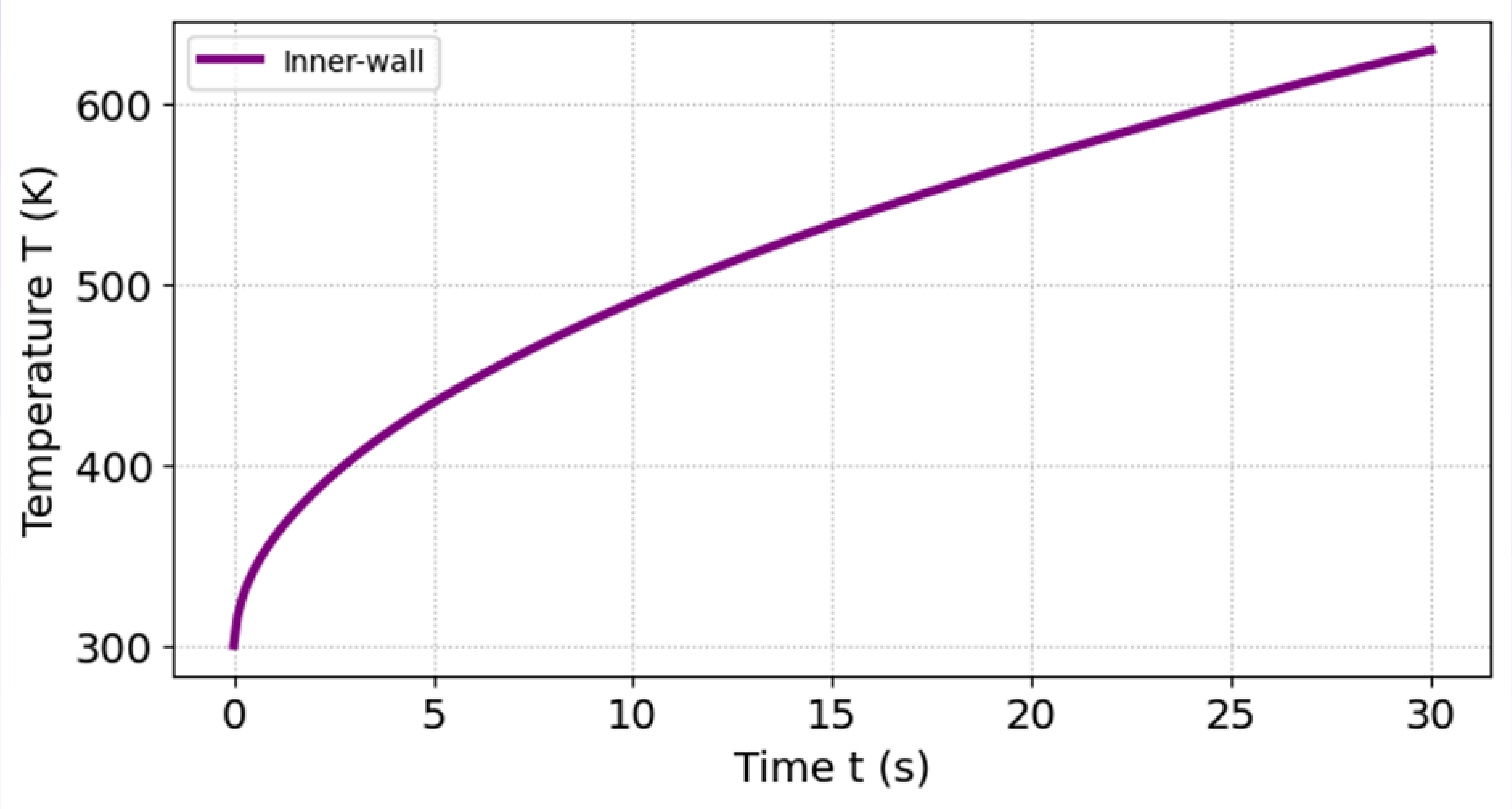
Inner wall Temperature: 630.087 K

Outer wall Temperature: 304.372 K



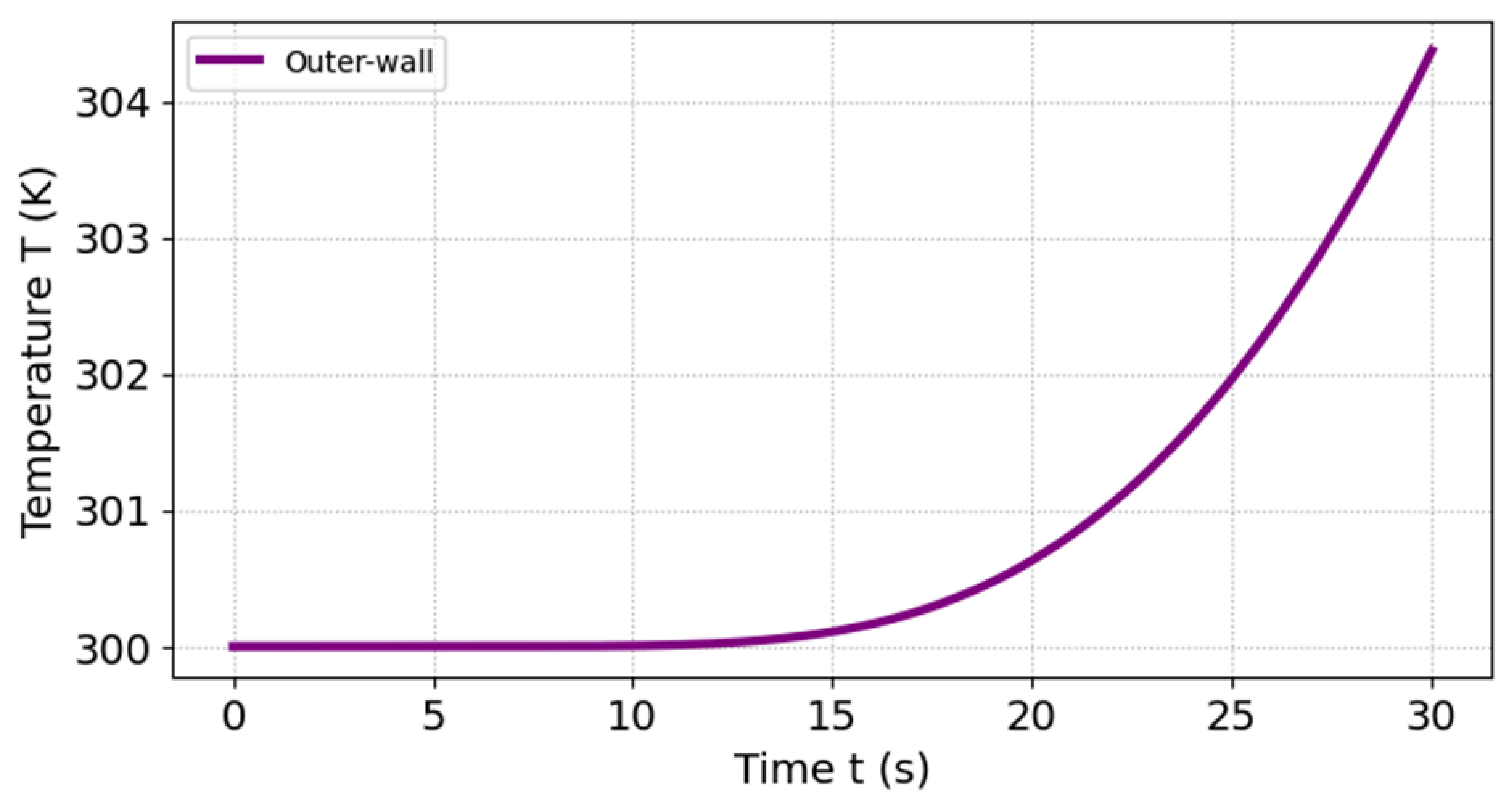
Results : Ansys APDL

Temperature variation of inner cell:



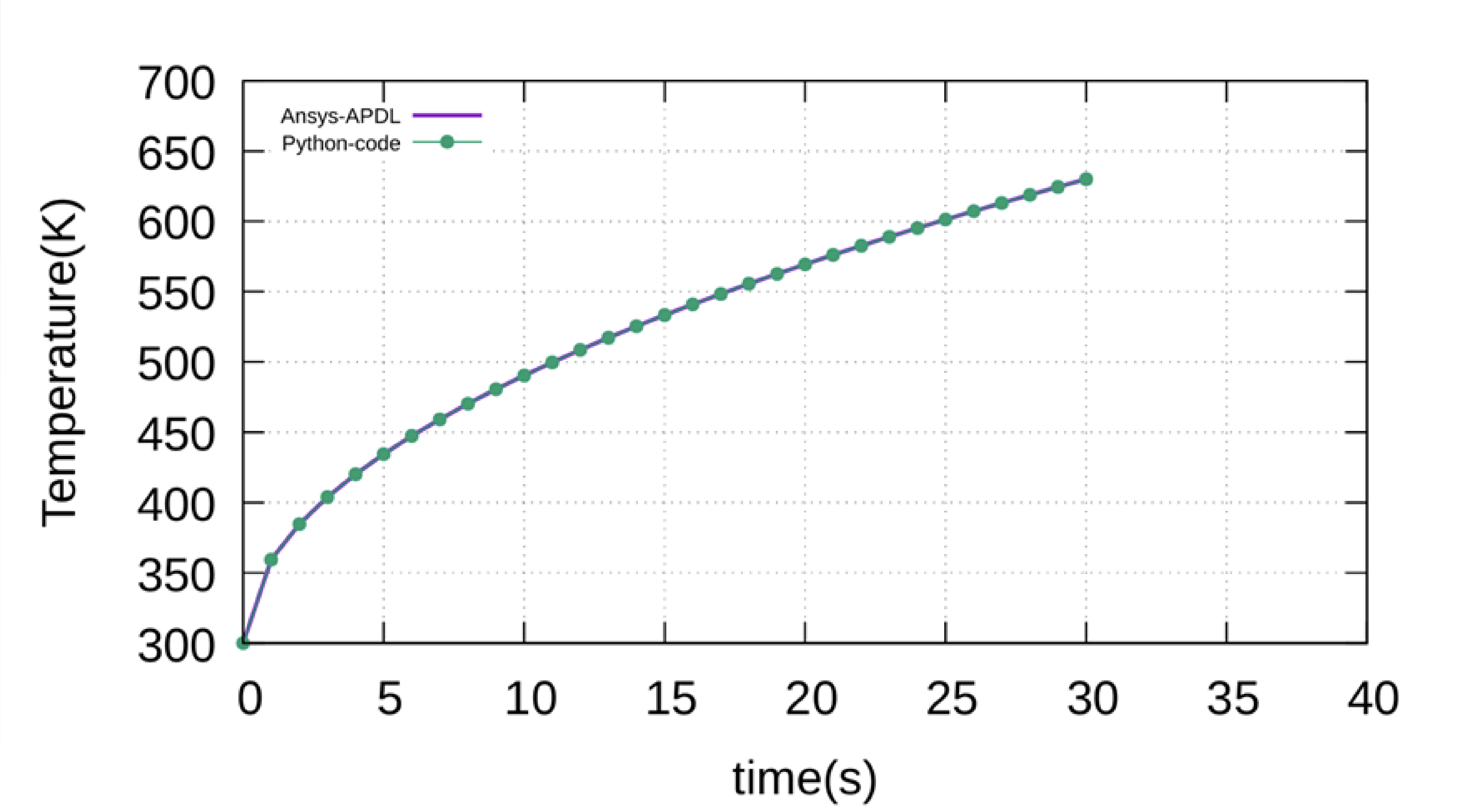
Results : Ansys APDL

Temperature variation of outer cell:



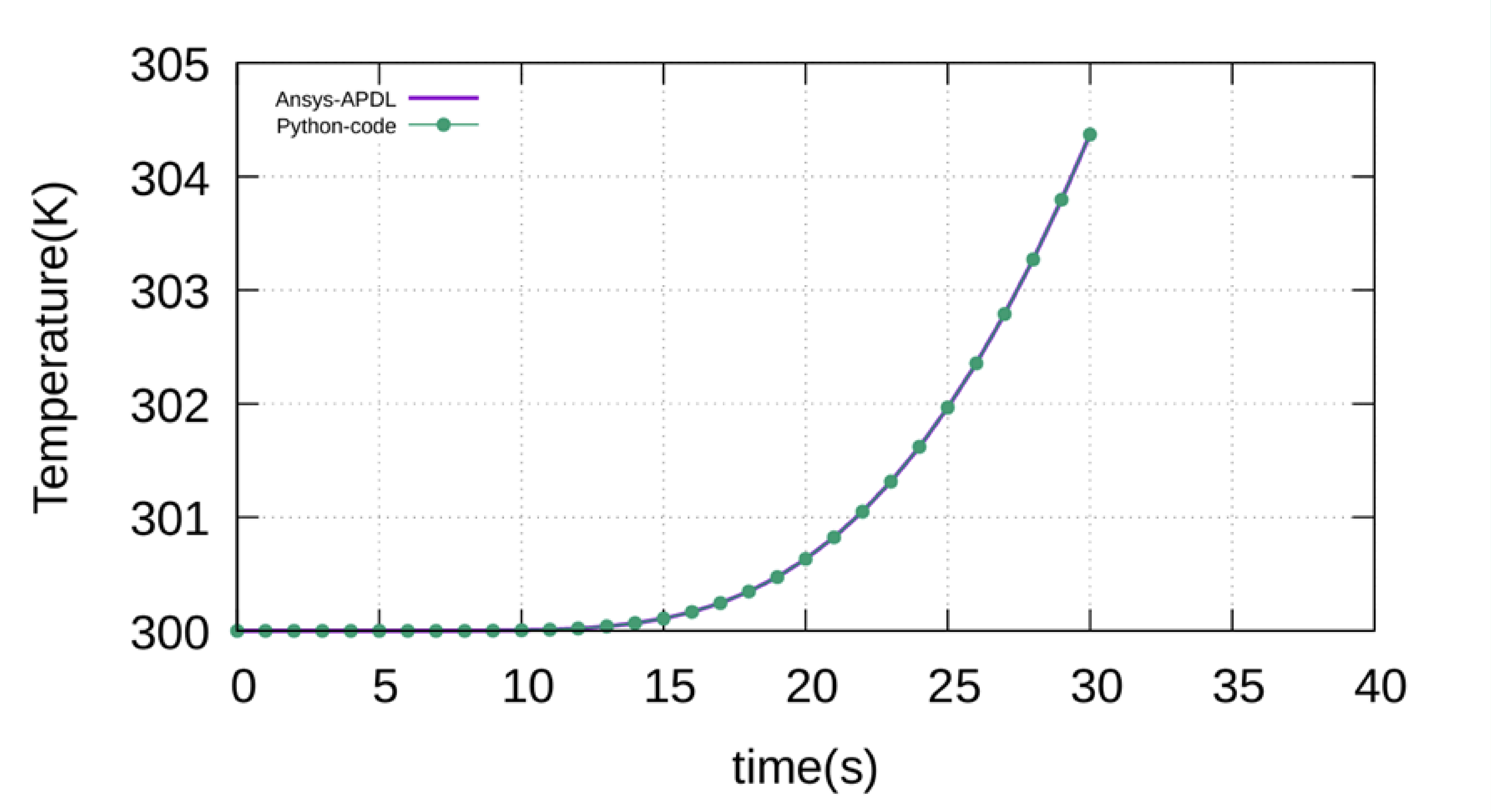
Verification

Inner cell:



Verification

Outer cell:



Technological Advancements

From structured mesh to unstructured mesh

FVM has evolved from structured meshes to unstructured/hybrid grids with higher-order schemes and adaptive refinement, enabling accurate modeling of complex geometries and flow features.

High-Performance Computing Power

HPC integration through parallelization, GPU acceleration, and domain decomposition allows FVM to solve large 3D, time-dependent problems with real-time analysis and multiphysics coupling.

Innovations & Integration

Hybridization with other methods, turbulence and multiphase models, plus machine learning and open-source tools make FVM more scalable, predictive, and widely accessible.

Thank you